



Figure 16-33 Urinary catheter placement in a male dog. **A**, The penis is extruded and the catheter is aseptically inserted. The catheter is passed until urine starts to flow. **B**, To anchor the catheter in the bladder, the balloon is inflated. **C**, For a closed collection system, a sterile extension set is aseptically connected to a collection bag.

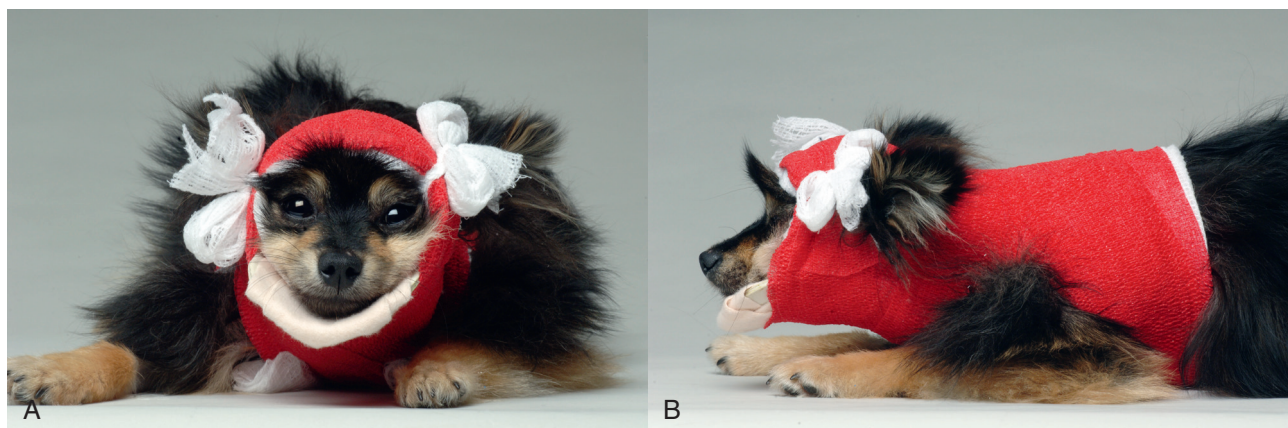


Figure 16-34 Coaptation for atlantoaxial luxation. **A**, Proper brace application to prevent the patient from ventral flexion of the neck. **B**, Lateral view of atlantoaxial brace. Note that the brace extends from the head to the cranial thorax.

to eat and drink. Many animals have difficulty eating, so syringe feedings with water and pureed food may be required. Keeping the patient in a comfortable position is very important. Positioning pillows and blankets around patients allows them to remain sternal, making it easier for them to eat, drink, and visualize the environment. For more severely affected cases, surgery is often required. Stabilization of the joint between C1 and C2 is performed. After surgical stabilization, a brace is still required for 4 to 6 weeks postoperatively.

Cervical spondylomyelopathy, also known as Wobbler's, is a condition primarily affecting large breed dogs. Clinical signs vary from ataxia to tetraplegia. Treatment options vary depending on the severity of the signs. Conservative management of mildly affected cases includes strict cage confinement for 4 to 6 weeks with only short walks outside with a harness. The use of neck leads or leashes should be avoided in cervical neurologic patients. Analgesics are administered as needed to control pain although many dogs with Wobbler's do not experience pain. Indications for surgery include substantial neurologic deficits or progression of signs despite conservative management. The surgical technique depends on the specific pathogenic changes identified on imaging, using MRI, CT, or myelography.

Postoperatively the cervical spinal cord patient can be very challenging. Frequently these patients are large to giant breed dogs that cannot be managed by the owner at home, requiring lengthy hospitalization until function returns. Early basic exercises focusing on activities of daily living are important for recovery. In severe cases patients may be so painful and weak that they are unable to lift their head. Foam supports and manual support or placement are required if the patient cannot maintain sternal recumbency. Nonambulatory tetraplegic patients should be assisted into sternal recumbency, or a standing position if capable, to eat and drink to prevent aspiration and regurgitation. Passive range of motion exercises to peripheral joints should be performed three to five times daily to prevent contracture and allow for joint nourishment. Assistive devices such as wheelchairs or a sling-lift system are frequently needed for postoperative cervical patients that are tetraparetic or tetraplegic (Figure 16-35).

Thoracolumbar Neurologic Patients

Intervertebral disk disease (IVDD) is the most common cause of thoracolumbar spinal injury in dogs. However, spinal tumors, fibrocartilaginous embolus, and trauma may also cause neurologic deficits. Hemilaminectomy to relieve the spinal cord compression is the most common surgical treatment of IVDD. Postoperative care of the IVDD patient focuses on pain control, bladder care, attention to bedding or housing, and postoperative rehabilitation. Often patient handling can be challenging if pain is not adequately controlled. In addition to oral and parenteral pain medications,



Figure 16-35 An assistive device, such as a Hoyer lift, can be used to manage a large tetraparetic or tetraplegic cervical patient.



Figure 16-36 An assistive device, such as an abdominal sling, can be used to help a thoracolumbar patient stand and ambulate.

cryotherapy, electrical stimulation, gentle massage, moist heat (not within 72 hours postoperatively), and therapeutic laser may provide additional pain control for the patient.

Often patients with paraparesis drag themselves with the fore limbs. It is essential to discourage this behavior, because it can cause severe skin wounds and abrasions, increase potential for urinary tract infections (especially in females), and place significant stress on the forelimbs. Proper fitting with a sling (Figure 16-36) in the immediate postoperative period is recommended for paraplegic and paraparetic patients. A wheelchair or cart (Figure 16-37) is indicated if the disability is expected to last longer than 4 to 6 weeks (potentially earlier in large and giant breed patients).

Summary

Nursing care of the rehabilitation patient is an essential part of the rehabilitation program. Several factors,



Figure 16-37 An assistive device, such as a cart, can be used to help a thoracolumbar patient stand and ambulate.

including the procedure and owner expectations, must be considered when caring for the rehabilitation patient in order to have a successful outcome. Whether the patient has had an orthopedic or neurologic injury or surgery, the ultimate goal is the same: to assist and care for the patient to achieve a functional recovery.

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17 Devices for Ambulation Assistance in Companion Animals

Denis J. Marcellin-Little and David Levine

Independent locomotion is critically important in dogs and cats. In fact companion animals that have lost the ability to walk independently are at risk of being euthanatized.¹ Many devices have been conceived to provide ambulation assistance and independence from owner care.² Humans use canes, crutches, braces, orthoses, and prostheses to assist with locomotion.^{3,4} In recent years the veterinary profession has been increasingly involved in physical rehabilitation for weak and paretic patients, and multiple devices aimed at providing ambulation assistance to animals have been created and used.^{2,5} The devices used to assist companion animals may support or protect weak, unstable joints or feet (orthoses, boots); replace a missing portion of a limb (prostheses); or facilitate locomotion (slings, carts, ramps, steps). Assistive devices may also decrease the complications present in recumbent patients, including decubitus ulcers.² The purpose of this chapter is to present the range of ambulation assistive devices and discuss their purpose, benefits, construction, and functional use for companion animals.

General Considerations When Using Assistive Devices for Ambulation

Assistive devices present unique challenges to rehabilitation clinicians because their design, fabrication, fitting, and the patient training are most often unique. Their successful use requires an accurate assessment of the biologic situation and the mechanical needs of the patient, requiring knowledge in the fields of orthopedics, rehabilitation, biomechanics, and prosthetics.⁶

Dogs wearing assistive devices should not be left unsupervised because they may damage their assistive devices or injure themselves. Assistive devices may be tolerated differently by individual animals. They may present risks when they are partially dislodged. As a general rule hair protects the skin from abrasions and hair should be left intact unless it interferes with the performance of an assistive device. In patients with long hair, the hair may be trimmed or clipped to enhance the fit of a device. New devices should be worn under controlled conditions (i.e., indoors and under direct supervision) and for limited periods of time, starting with a few minutes. Devices

should not interfere with blood flow. If a device is too tight blood flow to an extremity will be decreased, potentially leading to a loss of tissue oxygenation or impaired venous drainage with resultant swelling of the lower part of the limb. Devices should be stable on the limbs to avoid problems created by constant friction, slippage, or rotation. The condition of the limb should be assessed each time the device is removed. Skin redness, bruising, or focal hair loss should be noted early and immediately addressed, by potentially modifying the assistive device (removing hard material responsible for skin wear or adding foam protection). Devices should be inspected for wear and damage and cleaned daily. Cleaning is especially important if there are any incisions, wounds, or open sores in contact with the assistive device.

Boots

Boots are used to protect pads and feet from abrasion or to provide support. They may be used in normal dogs to protect against abrasions created by rough surfaces; for example, for search and rescue dogs working in areas with broken glass. Sled dogs may wear fleece boots to avoid the accumulation of ice on their feet or to avoid skin lesions from ice crystals, for example (Figure 17-1). Boots may be used to avoid abrasions of the dorsal surface of digits and metatarsal bones of dogs with sciatic nerve palsy, provided that they have some residual hock function. If the hock is flaccid, a support orthosis may also be necessary to maintain the lower limb in a functional standing position. Similarly boots may be used to protect the dorsal surface of the manus in patients with palsy of the branch of the radial nerve that innervates the extensor muscles of the antebrachium (Figure 17-2). Because most radial nerve palsies or brachial plexus injuries also affect the triceps brachii muscle and therefore severely limit weight bearing by the entire forelimb, boots may not provide adequate support alone. Boots may also be used in dogs with degenerative myelopathy and other conditions that lead to scuffing of the feet. Boots must be lightweight, particularly for patients that have weak motor function because the added weight of the boot on the foot may further decrease limb use. Boots may be made of nylon mesh, cotton, fleece,



Figure 17-1 Fleece boot made for a sled dog. The boot has a simple design, with a single hook-and-loop band used to secure it in place. These boots are placed on all feet during sled dog races. They protect the pads from abrasions and other injuries caused by snow and ice.



Figure 17-2 Golden retriever with a partial brachial plexus avulsion sustained as a result of a fall from a moving vehicle. A palsy of the radial nerve is present. Function of the triceps brachii is weak and antebrachial extensor function is absent. The dog is seen wearing a sock, a folded bandana taped on the caudal aspect of the carpus, and a support orthosis (*inset*) placed inside a boot, stabilized with athletic tape. Although gait abnormalities remain (the affected forelimb is placed under the chest by abducting and externally rotating his humerus at a walk), the dog is nevertheless able to bear weight on his affected limb.

neoprene, rubber, or other materials. They may have a rubber sole for increased traction.² They may be held in place through the use of straps or hook-and-loop fasteners. Bootstraps are most often placed immediately proximal to the distal portion of the metacarpal or metatarsal bones in order to secure the boot.

Orthoses, Including Sleeves and Braces

Orthoses are devices applied to limbs for support and protection. They are applied on limbs in order to assist or limit joint motion. The most common orthoses are used to restrict the excessive joint motion present in patients with ligamentous or tendinous problems, including patients with ruptures of their palmar fibrocartilage or common calcaneal tendon (*Figure 17-3*). Orthoses function by placing the limb segments proximal and distal to unstable joints in a foam-lined rigid shell. Orthoses are most often made of rigid thermoplastics. They may be lined with foam, fleece, or terry cloth. They are most often held in place with hook-and-loop fasteners. Orthoses may be commercially available premade or they may be custom made. Premade orthoses are affordable and are best suited to patients with normal anatomy and healthy skin (i.e., patients with rupture of the common calcaneal tendon). Custom orthoses are usually necessary in patients with abnormal anatomy (limb deformity, abnormal skin



Figure 17-3 Siberian husky with ectrodactyly (lobster claw deformity) of his right forelimb. The right carpus collapses when bearing weight (*upper inset*). A custom orthosis has been made that supports his carpus (*lower inset*). The stability of the limb within the orthosis could be optimized by placing a third strap across the distal aspect of the radius.

contours) and in very small or very large patients. They are generally based on a cast of the abnormal limb made by use of semirigid or rigid fiberglass or plaster of Paris. They may also be based on a computed tomography-based replica of the limb. Solid replicas may be made of cyanoacrylate-impregnated plaster, ABS plastic, or laser-cured photopolymers.⁷ Orthoses may be hinged to enhance their functionality.⁸ The hinges most often connect rigid shells placed around the pes (or manus) and the crus (or antebrachium). These hinges may be passive or dynamic (i.e., spring loaded). Passive hinges may be made of nylon or metal (Figure 17-4).⁸ Orthoses with passive hinges may enhance limb function because part or all of the motion of the tarsus (or carpus) may be maintained. Orthoses may have dynamic hinges that may be used to enhance the support of weak joints or to stretch contracted joints. Hinges are made dynamic by using springs embedded in the hinges or elastic bands.

Sleeves may be placed around limbs to provide protection or support, but they do not cover the feet. Although some sleeves offer protection of abnormal skin surfaces, for example in dogs with elbow hygromas or in recumbent patients at risk of developing decubitus ulcers, little is known about the benefits of sleeves for ambulation assistance. Sleeves may be reinforced, essentially becoming semirigid orthoses, by creating a pocket along their long axis and sliding a plastic or metal strip in that pocket.



Figure 17-4 Labrador retriever with a nonunion fracture of the metatarsal bones, hyperextended digits (*lower inset*), and a chronic nonhealing wound on the plantar surface of the foot. A hinged orthosis with a nylon hinge (*upper inset*) restores the dog's mobility.

Braces are placed across a joint, usually to limit its range of motion or promote its stability. Braces may be elastic or rigid. Rigid braces may also be used to modify the position of joints in nonfunctional limbs. This could be considered in patients with contractures, for example in patients with chronic grade IV medial patellar luxations that may have severe internal rotation of their tibia in relation to their femur. These patients may lose 60 degrees of stifle joint extension or more because of the shortening of their hamstring muscles as a result of the deformities and poor function of the quadriceps mechanism. A rigid brace may be considered in these patients with severe patella luxation and loss of stifle joint extension, when surgery is not a consideration or has failed (Figure 17-5). Braces may be made of foam-lined metal or thermoplastic, neoprene possibly reinforced with metal strips, or nylon. Most braces are hinged to allow joint motion. Hinged braces may be made of thermoplastic, neoprene, nylon, or other material.



Figure 17-5 A, Yorkshire terrier with medial patellar luxation, internal rotation of the tibia in relation to the femur and lack of stifle joint extension, making the leg nonfunctional. He has been fitted with a rigid orthosis (B, C). The orthosis matched the dimension of his leg, and is made of thermoplastic, foam, and three hook-and-loop straps fitting around the thigh, crus, and pes.

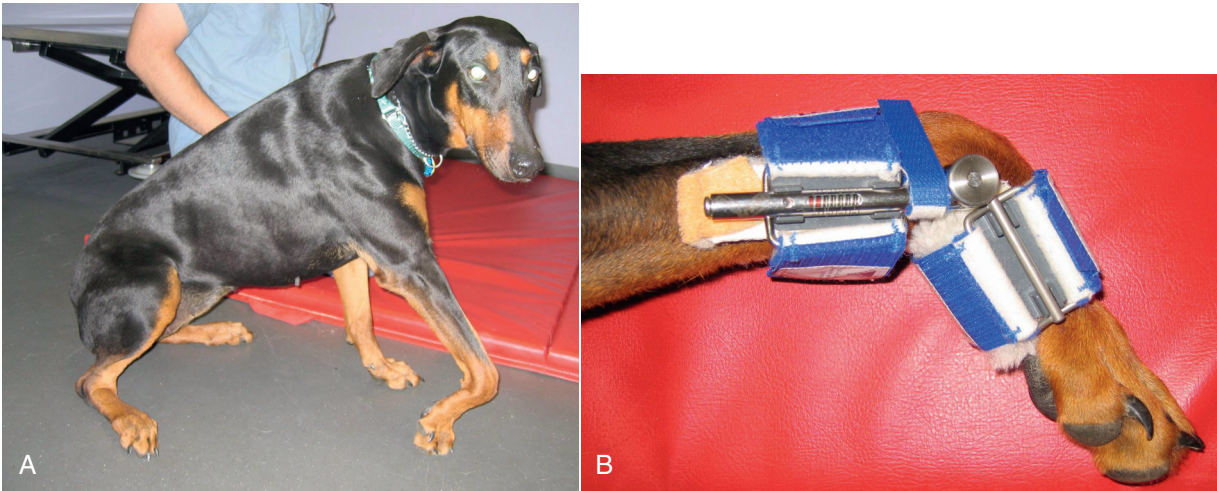


Figure 17-6 A, Doberman pinscher with an idiopathic loss of carpal extension. B, The antebrachial flexor muscles are stretched using a dynamic hinged brace that is worn for several hours at a time. The torque placed on the carpus is slowly increased over a period of weeks.

The hinges may be passive or dynamic (Figure 17-6). For example in the canine stifle joint, a brace theoretically could be used to decrease the stress placed on healing ligaments after traumatic stifle luxation, to limit the cranial thrust occurring while weight bearing in cranial cruciate deficient joints, or to improve the tracking of the patella. Despite efforts to design and commercialize several stifle joint braces, there is no brace that appears to predictably achieve any of these theoretical goals at this time. Knee braces are commonly used in humans for a variety of reasons, including protecting healing ligaments or improving patellar tracking.⁹ Functional knee braces do not prevent joint subluxation in humans.¹⁰ They appear to have a role in the anterior cruciate deficient knee, but only in conjunction with activity modification.¹⁰ One problem in stabilizing the cruciate ligament deficient stifle is the conformation of the rear limb, and the slope of the tibial plateau at a standing angle. To eliminate subluxation of the tibia relative to the femur, it would be necessary to completely compress the soft tissues of the femur and tibia with the splint, severely compromising these tissues.

Prostheses

Prostheses are external devices used in most instances to replace a missing manus or pes. Like custom orthoses, prostheses are often fabricated based on casts of the affected and opposite limbs. The layer of the prosthesis that contacts the skin, named *contact layer*, may be a silicone liner or foam adhering to a shell. Standard pediatric silicone liners may be used for long antebrachia. Custom silicone liners may be used for smaller or larger patients and for other body parts (i.e., thighs). Liners may be secured to shells using locking pins embedded in their distal portions or they may be held in place by friction within the

shell. Most prostheses have custom shells made of rigid thermomoldable plastic, fiberglass, or carbon fiber. Thermoplastics are cheapest, and their relative lack of strength and stiffness relative to other materials does not represent a significant drawback in smaller patients. Fiberglass or carbon fiber may be preferred for large patients. Little is known about the optimal shape of external prostheses for partially amputated forelimbs or pelvic limbs in companion animals.⁵ However, it is likely that a longer residual limb will provide better stability and reduce chances of skin breakdown. The stability of the prosthesis can be defined as axial, angular, and torsional. Axial stability is the displacement of the prosthesis along the long axis of the limb (i.e., the lack of movement of the prosthesis when it is being pulled on and pushed on). Axial stability is for the most part the consequence of the shape of the bone and surrounding soft tissues. It is also influenced by skin mobility in relation to underlying soft tissues. Axial stability often relies on the presence of bone prominences that are used to *lock* the prosthesis in place, creating *suspension*. Companion animals, as a general rule, have more discrete bone prominences than humans and the process of locking the prosthesis in place is harder to achieve. Angular stability is the lack of movement of the prosthesis in relation to the residual limb when mediolateral and craniocaudal forces are applied to it. Angular stability is most often the consequence of the length of a residual limb segment: the longer the better. Torsional stability is the lack of motion of the prosthesis in relation to the residual limb when internal or external torsion is applied. Torsional stability may be hard to achieve when a limb segment has a small diameter and is relatively round. This is particularly relevant in the antebrachium. Axial and torsional stability of prostheses are increased when the prostheses span a joint (i.e., the hock, stifle, or elbow joint). This is often



Figure 17-7 Sheltie who lost his toes after surgical resection of a soft tissue sarcoma of his thigh. The distal portion of his stump is sensitive to touch. His stump is placed in a silicone-lined glove finger and fitted in a hinged prosthesis held in place with four hook-and-loop fasteners.

achieved through the use of hinges placed over the center of rotation of the joint proximal to the end of the residual limb.

Adjustable devices may facilitate the function of the distal portion of these prostheses.⁷ The weight-bearing surface of prostheses is most often rounded and made of low-wear, nonskid rubber. The rubber stiffness is chosen based on patient size, mobility, and anticipated level of activity. Some prostheses are hinged at the carpus or tarsus. These hinges may enhance the stability of the prosthesis on the limb by increasing the contact area between the shell and the skin without excessively interfering with joint motion. Hinges may also be included in the prosthesis to promote a joint position more compatible with weight bearing. Hinges add complexity and cost to prostheses, but they may greatly enhance limb function (Figure 17-7).

Straps and Bands

Straps and bands may be used for ambulation assistance. Elastic bands designed to assist patients with compromised hock flexion, for example in patients with sciatic or peroneal nerve palsy, are commercially available (Figure 17-8). These elastic bands are attached to the mid-metatarsal region and to a ring on a chest harness. Elastic bands or tape may also be used to form hobbles, limiting forelimb or pelvic limb abduction. Pelvic limb hobbles are most often used in the management of patients recovering from ventral hip luxations. Forelimb hobbles are most often used to manage patients with injuries of the medial glenohumeral ligament.¹¹



Figure 17-8 Labrador retriever with a sciatic neuropathy, weak hock flexion, and scuffing of hind limb toes. Elastic bands within nylon sleeves are connecting his pes to his dorsal midline. The bands promote hock flexion and eliminate scuffing.



Figure 17-9 Dalmatian is supported by a chest and pelvic limb sling. The sling has a longitudinal chest handle and a transverse pelvic handle. It may be used to control the dog and to avoid falls.

Slings

Slings may be used to support weak patients with neurologic or musculoskeletal conditions. They may also be used to control dogs or avoid falls. Examples include patients recovering from orthopedic procedures of their pelvic limbs, such as total hip arthroplasty. Slings may be made of neoprene or other material, including cotton or nylon. Slings often have straps with an adjustable length for comfort of the therapist and should also be washable (Figure 17-9). Thin support sling bands may be used to control dogs recovering from injuries or surgery



Figure 17-10 Labradoodle with an amputation of his left pelvic limb. The dog is recovering from a total hip replacement of his right hip joint preformed to manage a femoral malunion with severe osteoarthritis. The dog is trotting on a treadmill, controlled and supported by a chest harness and a belly sling. The sling is made by wrapping self-adherent bandaging tape around a surgical gown.

(Figure 17-10). Slings supporting the abdomen may be used in conjunction with slings supporting the pelvic limbs and thoracic limbs. Subjectively abdominal slings do not appear to assist with locomotion as effectively as pelvic limb slings in dogs with pelvic limb weakness. Specialized slings are available to assist forelimb amputees (Appendix). Slings may be combined with exercise bands to improve the rehabilitation and locomotion of neurologic patients recovering from injuries or surgery. Patients who cross their pelvic limbs may benefit from the placement of a rolled cotton towel within a pelvic limb sling.²

Carts

Carts provide temporary or permanent assistance to non-ambulatory patients through support and prevention of the deleterious effects of recumbency. Several companies in North America have commercially available carts (Appendix—Manufacturers of Supplies and Equipment). Carts may be made of wide variety of materials. Their frames may be made of aluminum rods or tubing, or PVC piping.¹² Tires may be rigid or pneumatic. Wheels may be connected using ball bearings. As a consequence, the performance, weight, durability, and cost of ambulation carts are quite variable.

Paraparetic or paraplegic patients are usually fitted with carts that have two wheels or, possibly, three wheels if a small wheel is used on the caudal aspect of the cart. These paracarts support the lumbar spine and pelvic limbs by offering a rigid resting surface for the pubic and ischial symphyses or by slinging the pelvic limbs (Figure 17-11).



Figure 17-11 Saluki supported by a paracart. The pelvis is supported by a rigid base made of foam-covered tubing. The non-functional pelvic limbs are held in a slightly flexed position, off the ground. The dog's back is horizontal, and weight distribution between forelimbs and pelvic limbs appears optimal. (Courtesy of Mari Hiruma, Dog Will Project, Tokyo, Japan.)

Rigid resting surfaces usually are foam-covered aluminum tubing. Cart slings are most often made of neoprene with nylon straps and side release clips but they may be made of nylon mesh, cotton canvas, or other material. Optimal support materials are strong, breathable, washable, soft, and have some elasticity. Subjectively, rigid resting surfaces may be ideally suited to dogs without motor function in their pelvic limbs, and slings appear ideally suited to dogs with residual or intact motor function that need pelvic limb support. Dogs can spend several hours each day in properly fitted and balanced carts. Forelimb paracarts may also be used to support the chest of dogs with forelimb problems. This may be done using a special balanced two-wheeled cart or a quad cart (one with four wheels). Two-wheeled carts are more compact, more maneuverable, and cost less than quad carts. Tetraparetic or tetraplegic patients are usually fitted with “quad” carts. Like para carts, quad carts are available with rigid resting surfaces or slings.

Most carts are custom made. Cart measurements are specific to each manufacturer. The standing measurements required for cart constructions include: height of the dorsal midline, caudal to the scapula and at the lumbosacral joint; forelimb circumference proximal to the elbow joint; chest length and circumference, thigh length and circumference at the flank fold; height at the groin; width of the chest, abdomen, and pelvic regions; and width between the front legs. Some carts, however, are adjustable in height, width, and length.

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SECTION IV

Therapeutic Modalities

18 Superficial Thermal Modalities

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Thermotherapy is the use of superficial heat and cold as a therapeutic modality for the treatment of disease or trauma, and it may be applied using many different methods.

Superficial heat and cold have been used for centuries to manage soft tissue and joint injuries with the specific goals of relieving pain, altering the physiologic processes underlying tissue healing, and affecting the elasticity of connective tissue, including muscle, tendon, ligament, and joint capsule.¹ The primary goal of any thermal modality is to facilitate the ultimate therapeutic modality of exercise.

A dog adjusts its temperature based on the information supplied by the peripheral receptors, and, based upon the information, activates mechanisms such as trembling, sweating, and modifying the local circulation. Thermotherapy works on this latter process.¹

Thermotherapy applications may be simple or sophisticated, being delivered through hot or cold water, hot or cold packs, hot or cold whirlpools, ice massage, contrast baths, cold compression units, and other methods.¹

Before the application of heat or cold, the skin should be checked for the level of sensation, and if decreased or absent, extreme care must be used when applying thermal modalities to prevent injury. The skin should also be checked for any redness, burns, scabs, or wounds because these may alter the thermal transmission or affect the method of treatment. The area being treated should be cleaned to remove dirt or any other substance on the skin that could decrease conductance or alter the uniformity of the modality. The skin should also be inspected a short time after applying the modality to check for excessive temperature changes, and the skin should be monitored throughout the treatment. Cryotherapy is most effectively used immediately following trauma (accidental or following surgical procedures) to provide analgesia, control bleeding, and reduce inflammation, edema, and muscle spasm.

Cryokinetics combines cryotherapy with motion (passive, active-assisted, or active) to facilitate normal, pain-free movement, and to reduce edema through a muscle pumping action to facilitate the return of lymphatic fluid to the vascular system. The primary benefit of cryokinetics is to facilitate the patient's ability to perform pain-free exercise early in the rehabilitation process as long as the level of exercise remains below that which causes further injury. As signs of acute inflammation begin to subside, the clinician may elect to transition to the use of heat if indicated.

Superficial heat is best applied before performing stretches or exercises to improve range of motion (ROM). This is to take advantage of the benefits of tissue temperature elevation, including alteration of tissue viscoelastic properties, which results in increased soft tissue extensibility and decreased joint stiffness. In addition heat causes vasodilation, which increases tissue oxygenation and transport of metabolites to the exercising tissue, and increases the rate of enzymatic and biochemical reactions that may facilitate tissue healing. However, heat applied too early in the healing process may result in increased inflammation, increased enzymatic activity detrimental to cartilage (e.g., collagenase and gelatinase), and increased local metabolism.

The Physics of Thermotherapy

Thermotherapy applications fall in the infrared portion of the electromagnetic spectrum just beyond the wavelength and frequency ranges for visible light.² The shorter the wavelength, the greater the frequency and the more shallow the depth of penetration. The thermal electrophysical modalities discussed in this chapter fall within the category of “near infrared” and possess shorter wavelengths than electrical stimulating currents. Therefore, infrared electrophysical agents penetrate to a shallower depth compared with electrical stimulating currents. The biologic effects of

electromagnetic radiation depend on the frequency used, duration of exposure, tissue characteristics, and power density.²

Electromagnetic energy and its transmission through the body are governed by several physical laws. The first law is the Arndt-Schultz Principle¹: Tissue must absorb the energy produced by the thermal agent to stimulate the tissue's normal function. If the energy absorbed is insufficient to stimulate the tissue, there is no effect. If too much energy is absorbed, tissue damage may occur. The second law, related to the first, is the Law of Grotthus-Draper, which determines the fate of the energy. If a tissue does not absorb the energy, it is transmitted to deeper tissue layers. If the energy is absorbed more superficially, less energy is transmitted to deeper layers and less penetration of the energy occurs. When electromagnetic energy encounters the body's tissues, it has three possible fates: it may be reflected (e.g., reflected from the skin's surface), it may be refracted (e.g., at the interface between the dermis and subcutaneous fat), or it may be absorbed (e.g., by muscle). The estimated depth of penetration for most infrared thermal modalities (cold or hot packs, whirlpools, paraffin baths, or luminous infrared devices) is approximately 1 to 2 cm.

Heat may be transferred by the following mechanisms:

- **Conduction**—the transfer of heat by the direct interaction of the molecules in one area with those in another area (assuming the areas have a temperature difference). (Example: cold/hot packs)
- **Convection**—heat transfer by movement of air or fluid from a warm area to a cooler area. (Example: the circulatory system, air temperature)
- **Radiation**—the exchange of electromagnetic energy that occurs when there is a difference in temperature between two objects. Heat transfers from a warmer to cooler area. (Example: sun, infrared lamps)
- **Evaporation**—vapocoolant sprays are used to cool the skin and underlying tissues by causing evaporation and extracting heat from the tissues.
- **Conversion**—the conversion of a nonthermal form of energy (mechanical, chemical) that penetrates the tissues and is converted to heat. (Example: therapeutic ultrasound; sound waves being converted to thermal energy within the tissues)

Most thermal techniques transfer energy by conduction. Energy travels down a thermal gradient, with energy (heat) being removed from an object, rather than cold being added. Heat always travels from the warmer object to the cooler.³ According to Knight,⁴ the rate at which heat transfer occurs depends on the following factors:

- *The temperature difference between the body and the modality:* The greater the difference, the more quickly the transfer of heat occurs.
- *Regeneration of body heat and modality cooling:* As tissue gives up heat, the heat lost is replaced by circulating blood and surrounding tissues. Cold application tends to result in greater tissue temperature changes at greater tissue depths as compared with superficial heating modalities. For example, after a 20-minute ice pack application to the thigh muscles, the intramuscular temperature declined more slowly than the subcutaneous temperature, and there was less cooling at deeper tissue levels (and never reached as cold a temperature as the subcutaneous tissue). When the ice pack was removed the subcutaneous tissue temperature rose steeply, whereas the intramuscular temperature continued to decline for a period of time, possibly because heat was transferred from the deeper tissues to warm the superficial tissues, resulting in continued tissue temperature decreases at the greater tissue depths (Figure 18-1).
- *The heat storage capacity of the modality:* The more energy it takes to convert a solid to a liquid (e.g., ice to

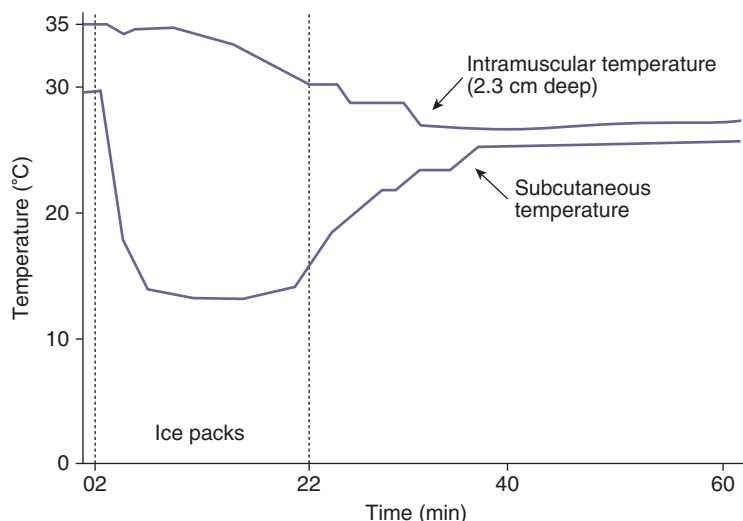


Figure 18-1 Intramuscular versus subcutaneous tissue temperature changes with cryotherapy. Intramuscular (gastrocnemius) and subcutaneous temperatures during and after a 20-minute ice pack application. Note that intramuscular temperature declined much more slowly than subcutaneous temperature and continued to decrease after the ice pack was removed. (Redrawn from Hartviksen K: Ice therapy in spasticity. *Acta Neurol Scand* 38(suppl 3):79-84, 1962.)

water), the greater its latent heat of fusion and its capacity to remove heat from a tissue. For example crushed ice packs require more energy for this conversion to occur as compared with semisolid gel packs. Therefore ice packs cool tissues for a longer time and to a greater extent than gel packs.

- *The size of the modality:* The larger the modality (e.g., large cold pack), the greater the energy storage capacity.
- *The area of the body in contact with the modality:* For example cold water immersion results in the greatest tissue temperature decline compared to other methods of cold delivery because a greater surface area is in contact with the cooling modality.⁵ Vapocoolant sprays result in the least temperature decline and provide only superficial cooling because of the relatively small surface area cooled.
- *The application duration:* The longer the contact time between the two surfaces, the more opportunity for energy exchange to occur.
- *Individual variability:* Individuals with lower body fat percentage exchange heat more quickly than those with greater amounts of subcutaneous fat.

Cryotherapy

Physiologic Effects of Cryotherapy

One of the oldest methods of physical therapy is cryotherapy, the application of cold after trauma or surgery. Cryotherapy is used not only during the acute phase of tissue injury and healing to mitigate the effects and sequelae of tissue injury, but also after exercise during rehabilitation to minimize adverse secondary inflammatory responses.⁵⁻⁷

The most important physiologic effects are:

- Vasoconstriction
- Decreased blood flow
- Decreased swelling
- Reduced enzyme-mediated tissue damage
- Analgesia

The decrease in blood flow slows edema formation after injury or after surgery. The cooling of the tissue reduces the metabolic rate of the treated tissue, decreases the rate of reactions related to the acute inflammatory process, causes the inhibition of enzymatic effects related to inflammation and minimizes the release of histamine, which reduces the tissue damage.^{6,7} At a temperature of 30°C or lower, cartilage-degrading enzymes (protease, hyaluronidase, collagenase) are inhibited. The vasoconstriction of blood vessels also may help to decrease pain by decreasing pressure on nociceptors, which may be stimulated as a result of increased tissue swelling and edema, as well as by other physiologic means such as decreased nerve conduction velocity.

The primary effect of cryotherapy is the vasoconstriction mediated by a local reflex and the central nervous system. The resulting reduced blood flow is useful not only to decrease swelling, but to control hemorrhage and the flow of inflammatory cells as well. The Hunting reaction, first described by Lewis in 1930, refers to the cyclical temperature oscillation of 2° to 6°C every 8 to 15 minutes after the tissue temperature approaches 2°C. This effect begins 20 to 40 minutes after cryotherapy application. He attributed these oscillations to cold-induced vasodilation (CIVD) as an attempt to protect tissues from cold-induced damage. However, his results have been incorrectly and widely applied to the sports medicine literature in an attempt to explain the success of cryokinetics.¹ Knight et al.⁸ replicated Lewis' experiment and concluded that Lewis' observation during cold immersion and subsequent rapid increase in temperature after cryotherapy are caused by temperature effects, not CIVD and subsequent increased blood flow. Based on these studies, the primary benefits of cryotherapy are twofold: decrease in cellular metabolism and analgesic effects to permit cryoexercise to facilitate rehabilitation.^{4,8}

The analgesic response to cryotherapy treatment is mediated by the peripheral nervous system. Peripheral thermal receptors are categorized as cutaneous myelinated type III or A delta, which are sensitive to pricking or sharp pain and cold; unmyelinated type IV or C receptors for aching pain; and unmyelinated type IV or C cutaneous receptors for pain and temperature (Table 18-1).

The unmyelinated cutaneous receptors respond to absolute temperature and the rate of temperature change.⁹ The cold-sensitive receptors begin firing at 36°C and increase their firing rate until reaching their maximum at approximately 25°C. The firing frequency drops sharply at temperatures below 20°C and is minimal by the time the receptors are cooled to 10° to 12°C. Conversely, the warm receptors also begin firing at 33° to 36°C, rapidly reach their maximum firing frequency at 43°C, and drop to a minimal firing rate at 45°C.^{9,10} Both the hot and cold receptors rapidly adapt to temperature changes.

Central thermosensitive neurons in the preoptic and anterior hypothalamus respond to temperature changes with autonomic responses to stimulate heat retention by the body, by altering cutaneous blood flow and thermoregulatory behavior such as panting. Figure 18-2 illustrates the regulation of temperature by the hypothalamus. The anterior hypothalamus also controls thermoinsensitive neurons, such as those that are sensitive to osmolarity and glucose concentration.^{9,11} Stimulating the posterior hypothalamus by heating does not result in the autonomic responses characteristic of the anterior hypothalamus, but behavioral responses to the heat load occur. Local warming of the medulla increases respiratory frequency. Autonomic thermoregulatory responses also take place at the spinal cord level.¹¹ The sympathetic nervous system response to